Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Computer Application

MCA | Sem-2 | Summer-2023

Course Code	:	2105CS202	**Date**	:	24-05-2023
Course Name	:	Python for Data Science	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Explain different coding styles available in python programming. **4**

(B) Explain different factors that control the speed of execution for data science application. **3**

OR

Explain the differences between conda and pip.

(C) List and explain any seven magic commands with an example. **7**

OR

Explain python ecosystem for data science.

Q.2 **(A)** Explain web scrapping using beautiful Soap. **4**

(B) Explain conditional selection in a numpy array. **3**

OR

Explain rand(), randint(), randn() functions of numpy with example.

(C) Explain numpy array creation functions with example. **7**

OR

Explain cross sections in DataFrame with example.

Q.3 **(A)** Explain time series with example. **4**

(B) Differentiate pandas vs numpy. **3**

OR

Differentiate join and merge.

(C) Consider following Restaurant_order.csv **7**

order_id	quantity	item_name	choice_description	item_price

Using pandas perform the following tasks.

1) Import the dataset from 'http://abc.com/ Restaurant_order.csv' and assign it to a variable called 'rsorder'
2) What is the number of observations in the dataset?
3) Which was the most-ordered item?

4) For the most-ordered item, how many items were ordered?
5) What was the most ordered item in the choice_description column?
6) How many items were ordered in total?
7) How many different items are sold?

OR

Consider following Salaries.csv

Id	EmployeeName	JobTitle	BasePay	TotalPay	Year

Using pandas perform the following tasks.

1) Read Salaries.csv as a dataframe called 'sal'.
2) Display first 25 entries of dataframe.
3) What is the average BasePay?
4) What job title does an "XYZ" employee hold?
5) What is the name of highest paid person?
6) What was the average (mean) BasePay of all employees per year? (2011-2014)?
7) How many Job Titles were represented by only one person in 2013?

Q.4 **(A)** Explain label, annotation and legend in matplotlib. **4**

(B) Explain choropleth graph with an example. **3**

OR

Explain Networkx library with an undirection graph.

(C) Write a python program read following marksheet.csv file and implement a heatmap on subject marks also handle missing data. **7**

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OR

Write a python program to implement countplot and boxplot with titanic dataset.

Q.5 **(A)** What is hashing trick? Explain countvectorizer with an example. **4**

(B) Explain bag of words model. **3**

OR

Explain generalization, overfitting, and underfitting.

(C) What is regression? Explain linear regression with an example. **7**

OR

What is classification? Explain k-Nearest neighbours with an example.

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